Part 4: Psychology of Testing-Bias Awareness Exercise

TC-001: Blank Password

Feature: Login with a Blank Password

User Story: Login with invalid login credentials should fail

Acceptance Criteria: Login fails with invalid credentials

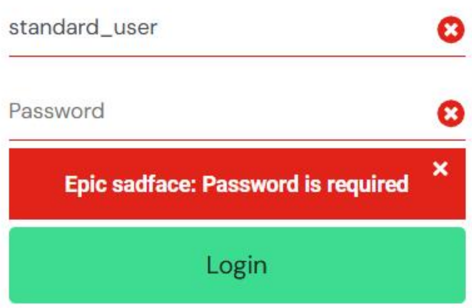
Steps to Reproduce:

Login with a valid user like standard_user and blank password

Expected Results:

Login fails and error message displayed

Status: Pass



TC002 Invalid username
Feature: Login – Invalid Username

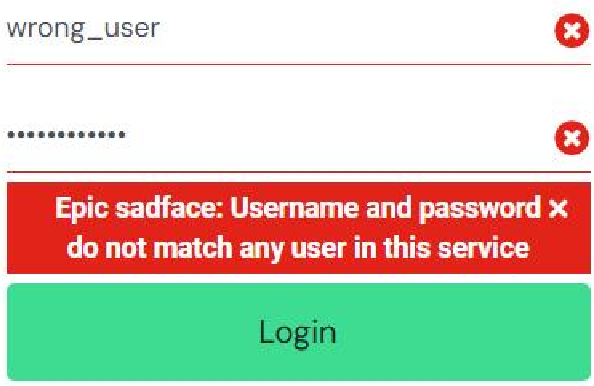
User Story: Login with invalid login credentials should fail

Acceptance Criteria: Login fails with invalid credentials

Steps to reproduce:
Login with username wrong_user and valid password secret_sauce

Expected Results:
Login fails and error message displayed

Status: Pass



TC003 Special Characters

Feature: Login – Special Characters

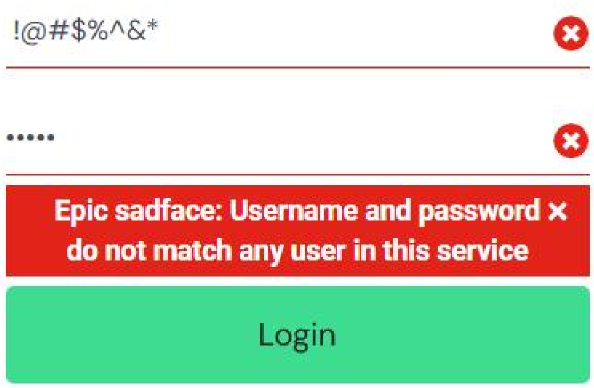
User Story: Login with invalid login credentials should fail

Acceptance Criteria: Login fails with invalid credentials

Steps to reproduce:
Login with username !@#$%^&* and password <>?/|

Expected Results:
Login fails and error message displayed

Status: Pass



TC004 : Both Username and Password Blank

Feature: Login – Blank Username and Password

User Story: Login with invalid login credentials should fail

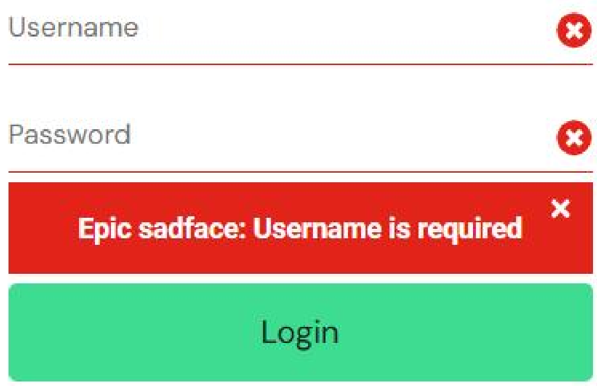
Acceptance Criteria: Login fails with invalid credentials

Steps to reproduce:

Login with both username and password left blank

Expected Results:
Login fails and validation error messages displayed for both fields

Status: Pass



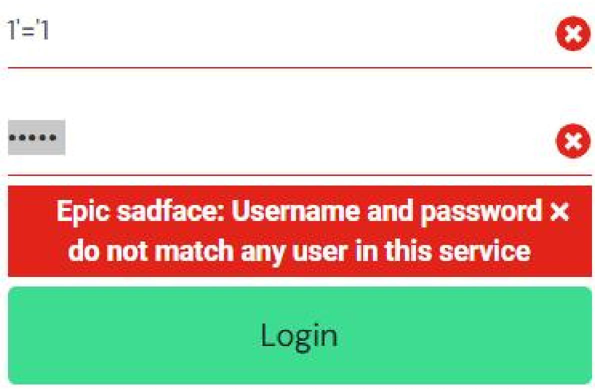
TC005 SQL Injection Attempt

Feature: Login – SQL Injection Attempt
User Story: Login with invalid login credentials should fail
Acceptance Criteria: Login fails with invalid credentials
Preconditions: None

Steps to reproduce:
Login with username 1'='1 and password ' OR '1'='1

Expected Results:
Login fails and system remains secure

Status: Pass



Reflection

Did you only test happy paths first?

No, I did not start with only happy path testing. I began with negative test cases following the activity instructions such as blank username/password, invalid credentials, and special character inputs to understand how the system behaves under invalid conditions first.

How would a developer vs tester approach differ?

A developer's mindset focuses on building solutions that meet business and user needs. Their energy is directed at solving problems and writing clean, efficient code.
Solution-oriented and optimistic:

- Focused on delivery and performance
- More inclined to see the system from a builder's point of view
- May unconsciously overlook defects due to confirmation bias

A tester's mindset involves questioning those solutions, identifying flaws, anticipating potential issues, and validating the built product.
Curiosity and critical thinking

- Attention to detail
- "Professional pessimism" - assuming things can go wrong
- Communication that is clear, diplomatic, and evidence-based
- A habit of asking: "What if this does not work?"

How confirmation bias can affect defect detection.

Confirmation bias is when we seek information that
supports what we already believe and shun information that disproves it. In software testing, this can
lead both developers and testers to overlook defects or risks.
For example, a developer may avoid writing negative test cases because they are confident the code
works. Similarly, a tester might over-test a feature from a developer they distrust, ignoring other
areas. Both patterns can lead to missed bugs.
To combat confirmation bias, testers must approach every feature neutrally, validate edge cases, and seek evidence that challenges assumptions. Testing is about discovering the unknown - not proving ourselves right.